Boot Controller FSM – Signal-Level Functional Description

FSM_RESET (Reset State)

In the reset state, the Boot Controller FSM is forced into a known and safe condition by the assertion of the global reset signal rst_n. During this state, the signal cpu_reset_n is kept deasserted to ensure that the processor remains inactive and cannot execute any instructions. The signal secure_mode is asserted to indicate that the system is operating in a restricted and secure condition. The boot status signal boot_done remains deasserted, signifying that the secure boot process has not yet begun or completed. No authentication-related signals such as start_auth are asserted in this state, and the FSM does not interact with any functional security modules. The reset state guarantees that the system starts from a fully secure and untrusted baseline.

FSM_IDLE (Idle State)

When the reset signal rst_n is deasserted, the FSM transitions into the idle state. In this state, the signal cpu_reset_n continues to remain deasserted, keeping the processor in reset, and the signal secure_mode remains asserted to maintain security restrictions. The signal boot_done stays low, indicating that no boot decision has been made. The FSM monitors the signal boot_req, which represents a system-level request to initiate secure boot. No control signals such as start_auth are asserted in this state, and no authentication or policy logic is triggered. The idle state functions as a controlled waiting phase until a valid boot request is received.

FSM_AUTH_START (Authentication Start State)

Upon detecting that the signal boot_req is asserted, the FSM transitions into the authentication start state. In this state, the FSM asserts the control signal start_auth to trigger the authentication engine to begin firmware verification. The signal cpu_reset_n remains deasserted to prevent processor execution, and secure_mode remains asserted to enforce security. The signal boot_done remains low, as authentication is still in progress. This state does not evaluate authentication results; instead, it serves as a synchronization point that cleanly initiates the authentication process by communicating with the authentication engine.

FSM_AUTH_WAIT (Authentication Wait State)

After initiating authentication, the FSM enters the authentication wait state. In this state, the signal start_auth is deasserted, indicating that the authentication trigger has been issued. The FSM keeps cpu_reset_n deasserted and secure_mode asserted, ensuring that the system remains inactive and secure while authentication is ongoing. The FSM continuously monitors the signal auth_done, which is asserted by the authentication engine upon completion of the verification process. No other control or decision signals are asserted in this state. The authentication wait state exists solely to allow the authentication engine to complete its operation without interference.

FSM_CHECK_RESULT (Authentication Result Check State)

When the signal auth_done is asserted, the FSM transitions into the authentication result check state. In this state, the FSM reads the signal auth_pass, which indicates whether the firmware authentication was successful or not. The signal cpu_reset_n remains deasserted, and secure_mode stays asserted while the decision is being

evaluated. The FSM does not assert boot_done yet, as the final action has not been taken. Based on the value of auth_pass, the FSM determines whether the system should proceed to a trusted boot or enforce a security failure response.

FSM_BOOT_ALLOW (Boot Allow State)

If the authentication result is successful (auth_pass = 1), the FSM enters the boot allow state. In this state, the FSM asserts the signal boot_done to indicate that the secure boot process has completed successfully and may assert a status signal such as boot_pass to reflect the positive outcome. The signal cpu_reset_n is asserted, releasing the processor from reset and allowing it to begin executing the authenticated firmware. Depending on system policy, the signal secure_mode may be deasserted to transition the system into normal operational mode. This state marks the successful establishment of hardware trust.

FSM_BOOT_DENY (Boot Deny State)

If the authentication result indicates failure (auth_pass = 0), the FSM transitions into the boot deny state. In this state, the signal cpu_reset_n remains deasserted to block processor execution, and secure_mode remains asserted to enforce security restrictions. The FSM asserts boot_done to indicate that the boot attempt has concluded unsuccessfully. The FSM may also rely on internal status signals such as retry indicators or failure counters, although it does not directly perform these functions. The boot deny state ensures that unverified firmware is never executed.

FSM_LOCKDOWN (Lockdown State)

When a critical security violation occurs, such as exceeding the allowed number of authentication failures, the FSM enters the lockdown state. In this state, the signal cpu_reset_n is permanently deasserted, ensuring that the processor can never execute. The signal secure_mode remains permanently asserted, and all boot-related signals such as boot_req are ignored. The FSM remains in this state indefinitely until a full power cycle or reset occurs. The lockdown state represents a terminal security condition designed to protect the system from persistent or severe attacks.

Conclusion

Across all FSM states, control signals such as cpu_reset_n, secure_mode, start_auth, boot_done, auth_done, and auth_pass are used to strictly regulate system behavior. The FSM ensures that firmware execution is permitted only after successful authentication and that any failure results in a secure and controlled response.